

Causal Survival - Case Base

2026-05-28

Imports

EDA

```
library("tableone")  
library("xtable")
```

Cox PH Model

```
library("survival")
```

SMD Balance

```
library("survey")  
library("cobalt")
```

Case Base

```
library("casebase")
```

EDA

Data Set

```
#set seed
set.seed(2026)

#re-factor the data set

#main variables of interest
time <- veteran$time
Delta <- veteran$status
A <- (veteran$trt - 1)

#covariates
CellType <- veteran$celltype
Karno <- veteran$karno
DiagTime <- veteran$diagtime
Age <- veteran$age
Prior <- (veteran$prior == 10)

#data set design
data.set <- data.frame(
  time = time,
  Delta = Delta,
  A = A,

  CellType = CellType,
  Karno = Karno,
  DiagTime = DiagTime,
  Age = Age,
  Prior = Prior
)
```

Table One

```

covariates = c(
  "CellType",
  "Karno",
  "DiagTime",
  "Age",
  "Prior"
)

table.one.original <- CreateTableOne(
  vars = covariates,
  strata = "A",
  data = data.set,
  test = FALSE
)

print(table.one.original, smd = TRUE)

```

		Stratified by A		
		0	1	SMD
##	n	69	68	
##	CellType (%)			0.465
##	squamous	15 (21.7)	20 (29.4)	
##	smallcell	30 (43.5)	18 (26.5)	
##	adeno	9 (13.0)	18 (26.5)	
##	large	15 (21.7)	12 (17.6)	
##	Karno (mean (SD))	59.20 (18.74)	57.93 (21.40)	0.063
##	DiagTime (mean (SD))	8.65 (8.76)	8.90 (12.27)	0.023
##	Age (mean (SD))	57.51 (10.81)	59.12 (10.28)	0.153
##	Prior = TRUE (%)	21 (30.4)	19 (27.9)	0.055

Propensity Scores

```
#generate propensity scores
propensity.score <- fitted(
  glm(
    A ~ CellType +
      Karno +
      DiagTime +
      Age +
      Prior,
    data = data.set,
    family = binomial(link = "logit")
  )
)

#separate propensity scores for exposed & unexposed groups
propensity.score.0 <- propensity.score[A == 0]
propensity.score.1 <- propensity.score[A == 1]

#visualize the propensity scores overlap
par(
  mar=c(2,2,2,0)
)

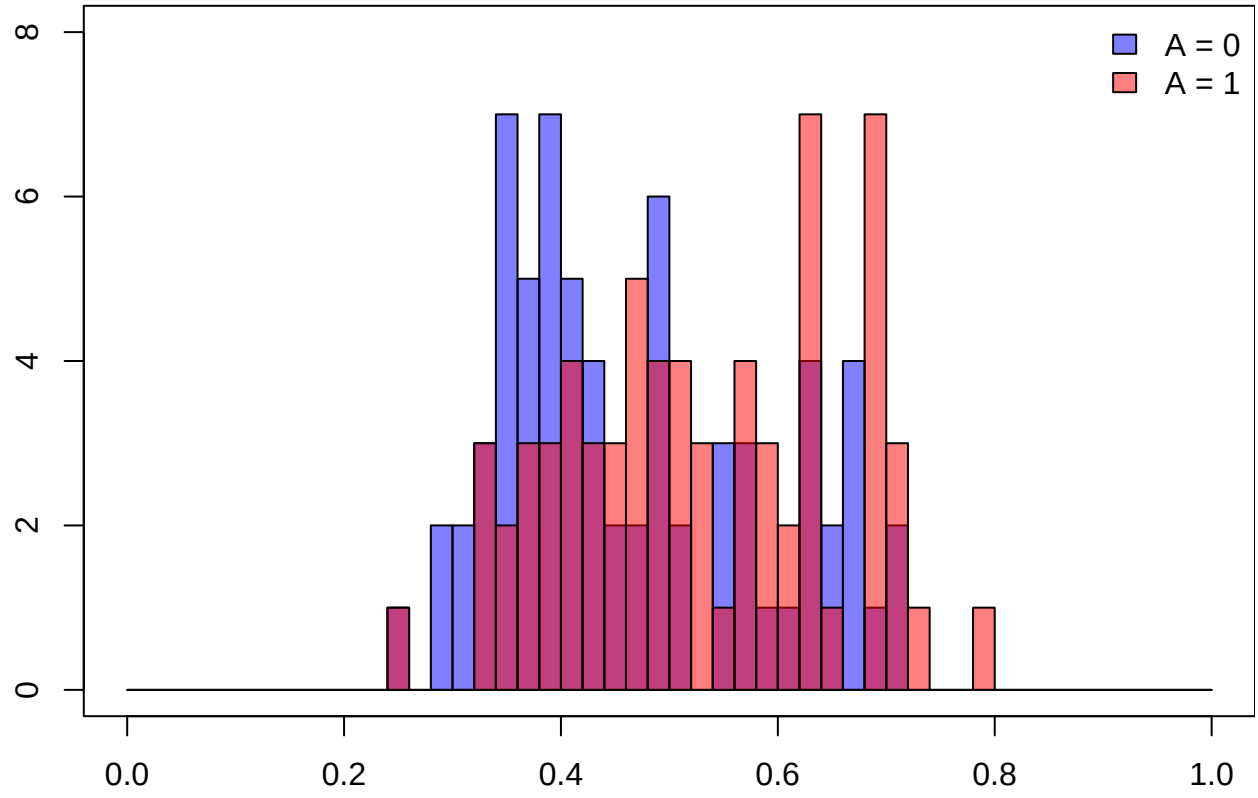
hist(
  propensity.score.0,
  col=rgb(0,0,1,0.5),
  breaks=seq(0,1,by=0.02),
  ylim=c(0,8),
  main="Propensity Score Overlap",
  xlab="Propensity Scores"
)

hist(
  propensity.score.1,
  col=rgb(1,0,0,0.5),
  breaks=seq(0,1,by=0.02),
  add=T
)

box()

legend(
  "topright",
  legend=c('A = 0', 'A = 1'),
  fill=c(rgb(0,0,1,0.5),rgb(1,0,0,0.5)),
  border="black", bty="n"
)
```

Propensity Score Overlap

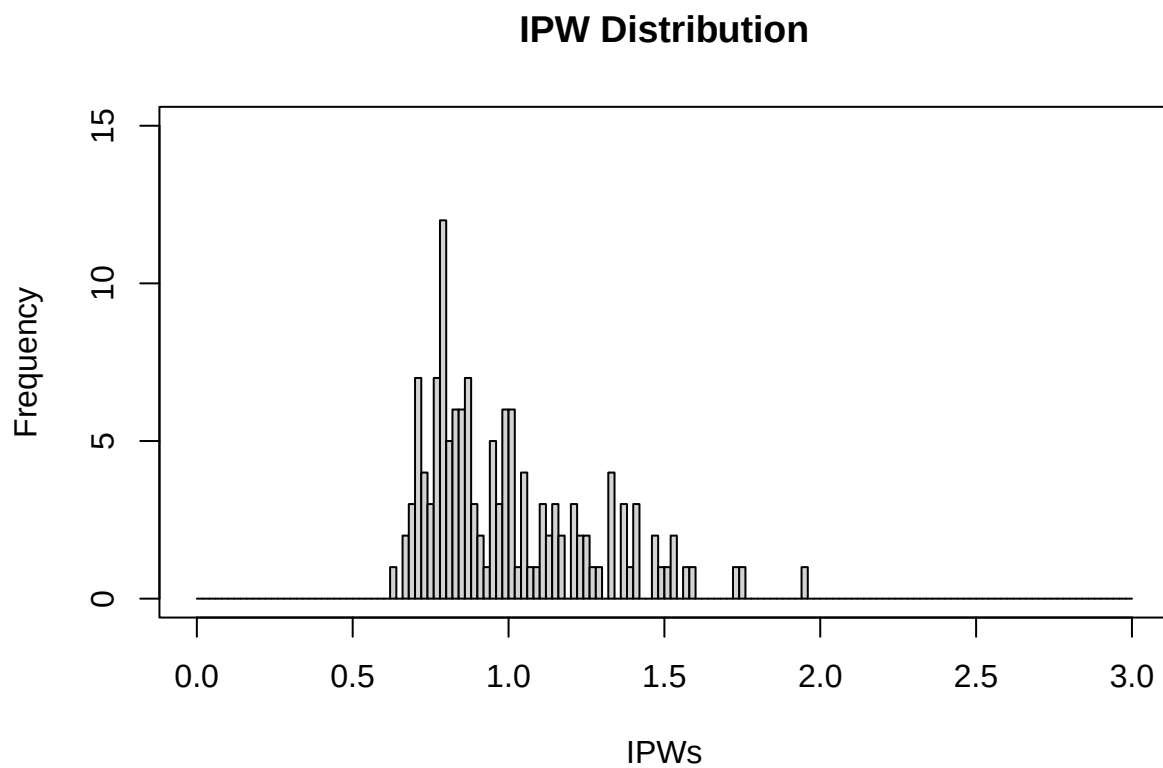


Stabilized IPW

```
p.A1 <- mean(A)

#calculate stabilized ipw
ipw <- ifelse(
  A == 1,
  p.A1 / propensity.score,
  (1 - p.A1) / (1 - propensity.score)
)

hist(
  ipw,
  breaks=seq(0,3,by=0.02),
  ylim = c(0, 15),
  main="IPW Distribution",
  xlab="IPWs"
)
box()
```



Covariate Balance

```
#calculate the covariate balance before & after IPW
balance <- bal.tab(
  A ~ CellType +
    Karno +
    DiagTime +
    Age +
    Prior,
  data = data.set,
  weights = ipw,
  method = "weighting",
  estimand = "Average Treatment Effect",
  un = TRUE
)
```

Note: 's.d.denom' not specified; assuming "pooled".

```
#visualize the covariate balance before & after IPW
love.plot(
  balance,
  stats = "mean.diffs",
  threshold = 0.1,
  abs = TRUE,
  line = TRUE,
  var.order = "unadjusted",
  colors = c("red", "blue"),
  shapes = c(17, 16),
  limits = c(m = c(0, 0.7)), #hard-coded 0.7??
  title = "Covariate Balance Before and After IPW",
  labels = TRUE
)
```

Warning: Standardized mean differences and raw mean differences are present in the same
plot. Use the 'stars' argument to distinguish between them and appropriately
label the x-axis. See 'love.plot()' for details.

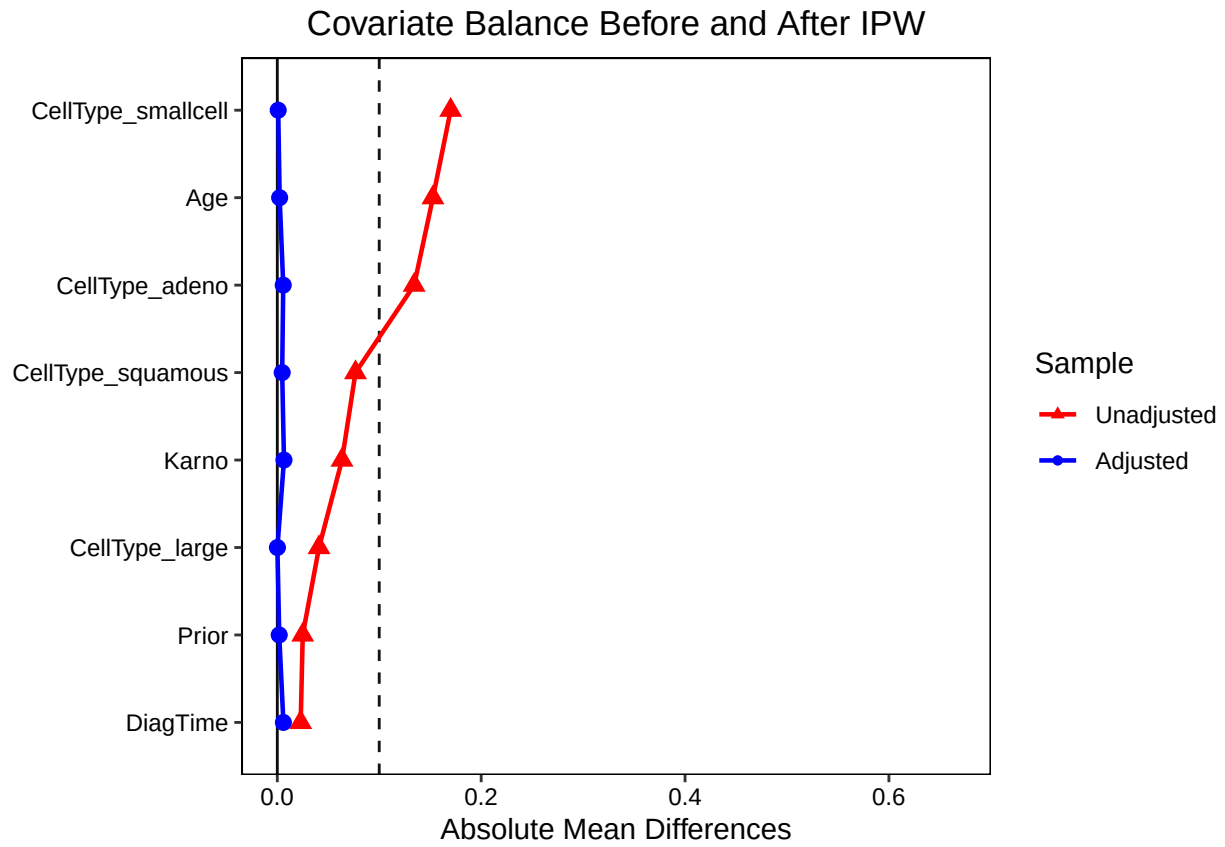


Table Two

```
data.set.ipw <- svydesign(
  ids = ~ 0,
  data = data.set,
  weights = ipw
)

table.one.ipw <- svyCreateTableOne(
  vars = covariates,
  strata = "A",
  data = data.set.ipw,
  test = FALSE
)

print(table.one.ipw, smd = TRUE)
```

##	Stratified by A		
##	0	1	SMD
## n	69.1	67.8	
## CellType (%)			0.016
## squamous	18.5 (26.8)	17.8 (26.3)	
## smallcell	23.9 (34.6)	23.4 (34.5)	
## adeno	13.2 (19.0)	13.3 (19.6)	
## large	13.5 (19.6)	13.3 (19.6)	
## Karno (mean (SD))	58.38 (19.12)	58.25 (21.01)	0.007
## DiagTime (mean (SD))	9.18 (9.78)	9.12 (12.33)	0.006
## Age (mean (SD))	58.30 (10.74)	58.32 (10.72)	0.002
## Prior = TRUE (%)	21.1 (30.5)	20.6 (30.4)	0.004

Estimation

Assumptions

Non-Informative Censoring

Model 1: Unadjusted Cox

```
#fit model
model.1 <- coxph(
  Surv(time, Delta) ~ A,
  data = data.set,
  robust = TRUE
)

#generate summary
summary(model.1)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A, data = data.set, robust = TRUE)
##
##      n= 137, number of events= 128
##
##      coef exp(coef) se(coef) robust se      z Pr(>|z|)
## A 0.01774  1.01790  0.18066  0.17664 0.1      0.92
##
##      exp(coef) exp(-coef) lower .95 upper .95
## A      1.018      0.9824      0.72      1.439
##
## Concordance= 0.525 (se = 0.026 )
## Likelihood ratio test= 0.01 on 1 df,  p=0.9
## Wald test              = 0.01 on 1 df,  p=0.9
## Score (logrank) test = 0.01 on 1 df,  p=0.9,  Robust = 0.01 p=0.9
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

Model 2: Covariate Adjusted Cox

```
#fit model
model.2 <- coxph(
  Surv(time, Delta) ~ A +
    CellType +
    Karno +
    DiagTime +
    Age +
    Prior,
  data = data.set,
  robust = TRUE
)

#generate summary
summary(model.2)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A + CellType + Karno + DiagTime +
##       Age + Prior, data = data.set, robust = TRUE)
##
##      n= 137, number of events= 128
##
##              coef exp(coef)    se(coef) robust se         z Pr(>|z|)
## A              2.946e-01  1.343e+00  2.075e-01  1.887e-01  1.561  0.11841
## CellTypesmallcell  8.616e-01  2.367e+00  2.753e-01  3.142e-01  2.742  0.00611
## CellTypeadeno     1.196e+00  3.307e+00  3.009e-01  2.760e-01  4.334  1.46e-05
## CellTypelarge     4.013e-01  1.494e+00  2.827e-01  2.524e-01  1.590  0.11188
## Karno            -3.282e-02  9.677e-01  5.508e-03  5.220e-03 -6.286  3.25e-10
## DiagTime          8.132e-05  1.000e+00  9.136e-03  7.948e-03  0.010  0.99184
## Age              -8.706e-03  9.913e-01  9.300e-03  1.030e-02 -0.845  0.39792
## PriorTRUE         7.159e-02  1.074e+00  2.323e-01  2.207e-01  0.324  0.74562
##
## A
## CellTypesmallcell **
## CellTypeadeno     ***
## CellTypelarge
## Karno             ***
## DiagTime
## Age
## PriorTRUE
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## A              1.3426      0.7448    0.9276    1.9433
## CellTypesmallcell  2.3669      0.4225    1.2785    4.3818
## CellTypeadeno     3.3071      0.3024    1.9255    5.6799
## CellTypelarge     1.4938      0.6695    0.9108    2.4498
## Karno             0.9677      1.0334    0.9579    0.9777
## DiagTime          1.0001      0.9999    0.9846    1.0158
## Age              0.9913      1.0087    0.9715    1.0115
## PriorTRUE         1.0742      0.9309    0.6970    1.6555
```

```
##
## Concordance= 0.736 (se = 0.021 )
## Likelihood ratio test= 62.1 on 8 df, p=2e-10
## Wald test = 68.69 on 8 df, p=9e-12
## Score (logrank) test = 66.74 on 8 df, p=2e-11, Robust = 52.32 p=1e-08
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

Model 3: Marginal Structural Cox

```
#fit model
model.3 <- coxph(
  Surv(time, Delta) ~ A,
  data = data.set,
  weights = ipw,
  robust = TRUE
)

#generate summary
summary(model.3)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A, data = data.set, weights = ipw,
##       robust = TRUE)
##
##      n= 137, number of events= 128
##
##              coef exp(coef)    se(coef)  robust se z Pr(>|z|)
## A -0.0000716  0.9999284  0.1812081  0.1789426  0      1
##
##      exp(coef) exp(-coef) lower .95 upper .95
## A    0.9999          1    0.7041      1.42
##
## Concordance= 0.477 (se = 0.027 )
## Likelihood ratio test= 0 on 1 df,  p=1
## Wald test               = 0 on 1 df,  p=1
## Score (logrank) test = 0 on 1 df,  p=1, Robust = 0 p=1
##
## (Note: the likelihood ratio and score tests assume independence of
## observations within a cluster, the Wald and robust score tests do not).
```

Model 4: IPW Case Base PH

```
#fit model
model.4 <- fitSmoothHazard(
  Delta ~ time * A,
  data = data.set,
  ratio = 10,
  weights = ipw
)

#generate summary
summary(model.4)
```

```
## Fitting smooth hazards with case-base sampling
##
## Sample size: 137
## Number of events: 128
## Number of base moments: 1280
## ----
##
## Call:
## fitSmoothHazard(formula = Delta ~ time * A, data = data.set,
##   ratio = 10, weights = ipw)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -4.8829486  0.1917106 -25.470   <2e-16 ***
## time         0.0004468  0.0012103   0.369   0.7120
## A            0.3243611  0.2563565   1.265   0.2058
## time:A       -0.0025597  0.0014604  -1.753   0.0796 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 857.86  on 1407  degrees of freedom
## Residual deviance: 848.75  on 1404  degrees of freedom
## AIC: 856.75
##
## Number of Fisher Scoring iterations: 6
```